

MATH 110

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What is a ODE? Recall that an ordinary differential equation (ODE) are equations for functions of a single variable, $u(t)$, $t \in \mathbb{R}$

$$\frac{du}{dt} = u(1 - u) \text{ is an ODE}$$

Here we call t the independent variable and u the state/dependent variable. In general, an ODE has the form $F(t) = (t, u, \frac{du}{dt}, \frac{d^2u}{dt^2}, \dots)$

Note that the state variable can have many components.

$$\mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} \in \mathbb{R}^2$$

then

$$\frac{d\mathbf{u}}{dt} = \mathbf{u} \text{ is an ODE}$$

What is a PDE? A PDE is an equation for a function with more than 1 independent variable, e.g. $u(x, y)$.

$$\begin{aligned}\text{Notation: } u_x &= \frac{\partial u}{\partial x} = \partial_x u \\ u_y &= \frac{\partial u}{\partial y} = \partial_y u \\ \text{etc.}\end{aligned}$$

Examples of PDEs

- Heat/diffusion equation for $u(x, t)$

$$u_t = k u_{xx} \quad k \text{ constant}$$

- Wave equation for $u(x, t)$

$$u_{tt} = c^2 u_{xx} \quad c \text{ constant}$$

- Laplace's equation for $u(x, y)$

$$u_{xx} + u_{yy} = 0$$

ex Consider the heat flow in a bar of length l

- $u(x, t)$ is the temperature of the bar at position x , time t
- Suppose the ends are held at temperature 0, and the initial temperature is $\phi(x)$
- The temperature evolves according to the heat equation $u_t = k u_{xx}$ where $t \geq 0, 0 \leq x \leq l, k$ constant.

We also have:

- Boundary conditions (BCs):

$$u(0, t) = u(l, t) = 0 \quad \forall t$$

- Initial conditions (ICs):

$$u(x, 0) = \phi(x)$$

Remarks

- The order of the PDE is the order of the highest derivative that occurs.
- The heat equation above is 2nd order
- A general 2nd order PDE for $u(x, t)$ is:

$$G(x, t, u, u_x, u_t, u_{xx}, u_{tt}, u_{xt}) = 0$$

- A PDE in the above form is linear if G is a linear function of u and all of its derivatives. It is nonlinear otherwise.
- A linear PDE is homogeneous if every term contains u or some derivative of u . It is nonhomogeneous/inhomogeneous otherwise.

ex Consider the linearity and homogeneity of the following equations.

1. $u_t + e^{x^2t}u_{xt} = 0$
is a linear and homogeneous PDE.
2. $u_t + 3xu_{xx} = \frac{t}{x}$
is a linear and inhomogeneous PDE because of the $\frac{t}{x}$ term.
3. $u_t + uu_x = 0$
is a nonlinear PDE because of the uu_x term.
4. $u_{tt} + u_x + \sin(u) = 0$
is nonlinear because of the $\sin(u)$ term.

Operating notation & linearity

We can write the heat equation $u_t - ku_{xx} = 0$ as $Lu = 0$ where

$$L = (\partial_t - k\partial_x^2)$$

We call L the differential operator. L acts on (smooth) functions $u(x, t)$ to produce a new function.

$$\begin{aligned} L(u) &= Lu = (\partial_t - k\partial_x^2)u \\ &= u_t - ku_{xx} \end{aligned}$$

L is a linear operator iff

$$\begin{aligned} L(u + v) &= Lu + Lv \\ \forall \text{ functions } u, v \end{aligned}$$

and

$$\begin{aligned} L(cu) &= cLu \\ \forall \text{ constants } c \in \mathbb{R} \\ \forall \text{ functions } u \end{aligned}$$

ex Check if the heat equation has a linear differential operator.

1.

$$\begin{aligned} L(u + v) &= (\partial_t + k\partial_x^2)(u + v) \\ &= u_t + ku_{xx} + v_t + kv_{xx} \\ &= Lu + Lv \quad \checkmark \end{aligned}$$

2.

$$\begin{aligned} L(cu) &= cu_t + ck u_{xx} \\ &= cLu \text{ for constant } c \quad \checkmark \end{aligned}$$

Remarks

- If L is a linear operator then $Lu = f$ is a linear PDE

$$\underbrace{u_t + ku_{xx}}_{Lu} = \underbrace{e^{x^2 t}}_{f(x,t)} \text{ is linear}$$

- If L is a linear operator then the PDE $Lu = f$ is
 - homogeneous if $f = 0$
 - inhomogeneous if $f \neq 0$
- A linear, homogeneous PDE $Lu = 0$ has the superposition principle

- If u_1, u_2 are solutions to the PDE then $au_1 + bu_2$ is a solution for constants a, b .

$$\begin{aligned} Lu_1 &= Lu_2 = 0 \\ \Rightarrow L(au_1 + bu_2) &= 0 \end{aligned}$$

- The distinction between linear & nonlinear PDEs is very important.

Solutions to PDEs

- A solution to a PDE is a function u in which satisfies the PDE and any BCs/ICs upon direct substitution.
- The general solution to an ODE involves arbitrary constants. The general solution to a PDE involves arbitrary functions.
- Typically, we are interested in specific solutions.

[ex] Consider the heat equation $u_t - u_{xx} = 0$.

$$\begin{aligned} u_1(x, t) &= x^2 + 2t \text{ is a soln} \\ \rightarrow \partial_t u_1 - \partial_x^2 u_1 &= 2 - 2 = 0 \\ u_2(x, t) &= e^{-t} \sin x \text{ is a soln} \\ \rightarrow \partial_t u_2 - \partial_x^2 u_2 &= -e^{-t} \sin x + e^{-t} \sin x = 0 \end{aligned}$$

[ex] Consider the equation $u_x = tx$ (1st order, linear, inhomogeneous)

$$\begin{aligned} Lu &= f \text{ with } L = \partial_x \\ f(x, t) &= tx \end{aligned}$$

We can solve this by direct integration on both sides wrt x .

$$u(x, t) = \frac{1}{2}tx^2 + A(t)$$

Where $A(t)$ is an arbitrary function of t

[ex] Consider the equation $u_{tt} - 4u = 0$ (2nd order, linear, homogeneous)

$$\begin{aligned} Lu &= f \\ \rightarrow L &= (\partial_t^2 - 4) \\ f(x, t) &= 0 \end{aligned}$$

There is no explicit x -dependence

- We can treat this as an ODE with x as a parameter.
- 'Constants' depend on x
- General solution:

$$u(x, t) = A(x)e^{-2t} + B(x)e^{2t}$$